Pre-Lab 6: Radiometer

Before your own section, submit one Word/PDF on Canvas with Q1–Q6 answers. Run all supplied notebook cells. Include the two original figures and the two required molecular figures. Use the notebook's download pane to save PNGs. All supplied numerical states are illustrative.

Q1. From frames to rpm

Consecutive events occur at frames 0, 250, 500, ... , 2000. Actual capture rate is 1000 fps; one full revolution contains four verified events. Calculate elapsed time, revolutions, mean rpm and angular speed [rad/s]. Show the calculation and include the cumulative-revolutions figure.

Q2. Two measurement mistakes

Recalculate Q1 using a 60 fps playback rate instead of capture rate. Then use p = 1 with the true 1000 fps. Explain each error factor and how you will verify p in a real movie.

Q3. A simple rotational model

Run the model with final speed 90 rpm and time constants 1, 3 and 6 s. Include the response figure. Which approaches its final value more slowly? What fraction is reached at one time constant from rest? Derive the exponential solution from the linear rotational ODE.

$$I\frac{d\omega}{dt} = \tau_{\text{gas}} - \tau_{\text{loss}}$$

I is rotor inertia [kg·m²], ω is angular speed [rad/s], t is time [s], and the gas and loss torques are in N·m. For loss bω, b has units N·m·s and the mechanical time constant is I/b.

Q3 continued: assumptions

$$\omega(t) = \omega_\infty + (\omega_0 - \omega_\infty)e^{-t/T_m}$$

Subscripts 0 and ∞ denote initial and final speed. This solution assumes constant gas torque and constant linear loss. Vane heating can change torque, so an experimental time scale may combine thermal and mechanical effects.

Q4. Physics and limits

Explain why residual gas matters. Does increased light imply a known torque increase? Can lux from different lamp spectra be treated as irradiance in W/m²? Explain briefly.

Q5. Measurement plan

List what you will hold fixed, vary and record. Explain how to detect a changing speed and a missed crossing. Describe an industrial measurement using passage timing. Plan comparable initial states and a common observation window.

Q6. Required molecular Monte Carlo

Run the molecular section of the same notebook. It samples wall-crossing velocities, calculates collision impulses, averages the hot/cold pressure difference and compares with an analytical free-molecular wall model. No programming from scratch is required.

Q6a: include the hot/cold outgoing-speed cumulative-distribution figure. Explain the temperature effect and why wall-crossing samples are flux-weighted.

Q6b: include sampled and analytical normalized pressure difference versus wall-temperature contrast. Discuss equal temperatures, reversed hot/cold assignment, and diffuse fractions 1, 0.5 and 0.

Q6c: run the sample-count plot and explain how scatter changes with N. Explain why four times as many independent samples typically halves sampling error. This extra plot need not be inserted into the submission.

Q6d: compare the two illustrative thermal states in the notebook. Explain how both faces can become warmer while the driving temperature contrast decreases. Do not treat those temperatures as measurements of the bulb.

The molecular algorithm and model boundary

1. Choose N, reservoir temperature, wall temperatures and diffuse-reflection fraction. 2. Draw uniform random numbers and convert to flux-weighted normal velocities. 3. Compute wall impulse from incident and outgoing momentum. 4. Average and compare with the analytical expectation.

The molecular mass is illustrative nitrogen; wall temperatures and reflection probabilities are prescribed. The model has no gas–gas collisions, finite-vane edges, enclosure, heat balance or rotor dynamics. It is not a full DSMC calculation of the apparatus.

At equal wall temperatures the expected thermal pressure difference is zero; finite Monte Carlo samples can fluctuate around it. A fully specular wall does not exchange thermal energy in this model. Optical color does not determine molecular reflection probability.

After the experiment: required report connection

Use your measured rpm/lux and early/late plots together with the molecular-force figure. Relate gas momentum transfer to the observed rotation and possible changes in thermal contrast. Explain what the data support and what remains unmeasured. The normalized wall-pressure curve is not a calibrated rpm prediction.

Complete this connection in the lab report after collecting data. It is not a request for experimental measurements in the pre-lab submission.

Submission checklist

Q1 calculation and figure; Q2 error factors; Q3 derivation and response figure; Q4 physical explanation; Q5 test plan; Q6a–Q6d answers with two molecular figures. Four figures total. Keep explanations focused and define variables used in equations.